Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

P3 Particle Model of Matter — Revision Topics

P3.1 · Changes of state and the particle model — density, states, changes of state

- **Density.** How much mass fits in each unit of volume: $\rho = m/V$, in kg/m^3. Mass is conserved when a substance changes state, so only the volume (and so the density) changes.
- **RP17, measuring density.** Regular solid: mass on a balance, volume from its dimensions. Irregular solid: volume by displacement (displacement can, or final minus initial reading in a measuring cylinder).
- **Solid:** particles in a regular pattern, touching, vibrating about fixed positions. **Liquid:** touching, irregular arrangement, free to move past each other. **Gas:** far apart, random motion in all directions at a range of speeds. Close-packed solids and liquids have a high density; widely spaced gas particles give a low density.
- **Changes of state:** melting, freezing, boiling, evaporating, condensing, sublimating. All are physical changes: the material recovers its original properties when the change is reversed.

P3.2 · Internal energy and energy transfers — heating, specific heat capacity, latent heat

- **Internal energy** = the total kinetic energy plus the total potential energy of all the particles. Heating raises it, which either raises the temperature or produces a change of state.
- **Specific heat capacity** c: the energy to raise the temperature of 1 kg of a substance by 1 °C. The rise in temperature depends on the mass, the material and the energy input.
- **Specific latent heat** L: the energy to change the state of 1 kg with no change in temperature. *Fusion* = solid to liquid; *vaporisation* = liquid to vapour.
- **Heating and cooling graphs.** *Sloping sections:* temperature changes, E_k increases, E_p constant, use $\Delta E = mc\Delta\theta$. *Flat sections:* temperature constant, E_k constant, E_p increases, use $E = mL$. Multi-stage problems: one equation per section, then add.

P3.3 · Particle model and pressure — gases

- Gas molecules move in constant random motion. Temperature is related to their *average* kinetic energy.
- **Constant volume, temperature up:** the molecules move faster, hit the walls more often and with more force, so the pressure rises. Temperature down: the pressure falls.
- Four-link chain for a full answer: temperature → average kinetic energy → collisions with the wall (more frequent, each harder) → pressure.

Equations — units: mass in kg, energy in J, temperature change in °C

$$\rho = \frac{m}{V}$$

density = mass ÷ volume

ρ *in* kg/m^3, m *in* kg, V *in* m^3

Recall

$$\Delta E = m\,c\,\Delta\theta$$

change in thermal energy = mass × specific heat capacity × temperature change

c *in* J/kg °C

On the equation sheet

$$E = m\,L$$

thermal energy for a change of state = mass × specific latent heat

L *in* J/kg

On the equation sheet